

# Michał Siniarski

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## Personal Profile

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I am a graduate of Warsaw University of Technology and a dedicated AI Engineer currently working at a leading Polish scientific institute. My specialization lies in Artificial Intelligence, with a focus on a wide array of applications and methodologies. I thrive on exploring the multiple facets of AI and its real-world applications, constantly seeking new challenges and opportunities for growth.

## Experience

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### Air Force Institute of Technology | AI Engineer

Warsaw, Poland  
July 2024 - Present

- Management of AI team workflows, including task prioritization and strategic problem-solving to meet project milestones,
- Research and evaluation of state-of-the-art methods to select effective solutions across end-to-end ML pipelines,
- Design and implementation of ML architectures and preprocessing pipelines for multi-modal data (audio, visual, point cloud),
- Performance optimization through model training and evaluation cycles,
- Automation of development and research workflows using AI agents, including Codex and Claude Code, with multi-agent orchestration for parallel task execution.

### Key achievements and projects:

- CNN and ResNet based models for frequency domain signals (FFT) and spectrograms (STFT) analysis, SAE model for sound denoising,
- RandLANet model implementation for large scale point cloud semantic segmentation (Mean Intersection over Union at 87%),
- Optuna library based system for fast and efficient search of optimum model configuration, and training hyperparameters.

### Air Force Institute of Technology | Project Manager

Warsaw, Poland,  
April 2023 - June 2024

Management of team responsible for tasks within the European HYDEF defense program. Coordination of internal activities, in the field of direct-hit warhead technology evaluation. Communication with members of the project consortium and alignment with the overarching project objectives.

### Core contributions:

- Planning FEM simulations for design evaluation,
- Duties execution through PRINCE2 methodology,
- Working in an international environment,
- Presenting work plans and results to diverse international audiences,
- Work package Leadership.

### Air Force Institute of Technology | Internship

Warsaw, Poland,  
July 2022

Development of a Python-based system for Non-Destructive Testing (NDT) inspections. Extensive study and implementation of NDT methods in data acquisition, processing, and visualization for defect detection in aerospace components.

## Education

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- MSc**    **Warsaw University of Technology, Faculty of Power and Aeronautical Engineering**    2023 - 2025  
**Warsaw University of Technology:**  
**Aeronautical Engineering**
- Specialization: Automation and Aerospace Systems,
  - Master's thesis focused on the implementation of a Structural Health Monitoring (SHM) system, utilizing AI for fast direct hit parameters estimation,
  - Graduated with distinction.
- Eng**    **Warsaw University of Technology, Faculty of Power and Aeronautical Engineering**    2019 - 2023  
**Warsaw University of Technology:**  
**Aeronautical Engineering**
- Specialization: Aircraft Propulsion,
  - Engineering Thesis: Development of a non-destructive testing (NDT) station for aircraft inspection, involving hardware selection and full, open-source software implementation.

## Publications

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- Development of an Open-Source Robotic NDT Solution for Automated Composite Repair Testing**    Oct 2024
- Michał Siniarski*, Piotr Synaszko, Krzysztof Dragan
- [doi.org/10.2478/fas-2023-0009](https://doi.org/10.2478/fas-2023-0009) 

## Skills and technologies

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### Tech stack:

- **Python:** proficiency in libraries for processing, analysis, data representation and machine learning (**PyTorch**, **TensorFlow**, **scikit-learn**, **OpenCV**, **NumPy**, **pandas**, **Matplotlib**, **Optuna**, experience with field related tools (e. g. **Conda**, **UV**),
- **SQL:** intermediate knowledge,
- **Docker:** intermediate knowledge,
- **Rust:** basic knowledge,
- **MATLAB:** intermediate knowledge,
- **LaTeX:** intermediate knowledge,

### Others:

- Proficiency with **GitHub**,
- Familiarity with Linux OS and MacOS,
- NX Modeling – Certified by the Center for Competence and Training for modeling and assembly of 3D objects or structures.

### Spoken Languages:

- Polish (native),
- English (fluent).

## Interests

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- Programming,
- PC building,
- Photography.

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